Exam 3 Review

Math 742

2026-04-20

1. Bootstrap, Jackknife, Bias Correction

Key Notation (consistent with lecture)

- Data: $x_1, \ldots, x_n \sim F$
- Empirical distribution: $\hat{F}_n$
- Parameter: θ
- Estimator: $\hat{\theta} = T(x_1, \ldots, x_n)$
- Bootstrap samples: $x_1^*, \ldots, x_n^* \sim \hat{F}_n$
- Bootstrap estimator: $\hat{\theta}^*$

Bootstrap

Principle:

$$\hat{F}_n \approx F \quad \Rightarrow \quad \mathcal{L}(\hat{\theta}) \approx \mathcal{L}(\hat{\theta}^*)$$

Bias estimate:

$$\widehat{\text{Bias}}_{boot} = \mathbb{E}^*(\hat{\theta}^*) - \hat{\theta}$$

Mini Example

Let $\hat{\theta} = \bar{X}$. Then:

$$\widehat{\text{Bias}}_{boot} \approx \bar{T}^* - \bar{X}$$

Jackknife

Define leave-one-out estimator:

$$\hat{\theta}_{(i)} = T(x_1, \ldots, x_{i-1}, x_{i+1}, \ldots, x_n)$$

Jackknife estimator:

$$\hat{\theta}_{jack} = \frac{1}{n} \sum_{i=1}^{n} \hat{\theta}_{(i)}$$

Bias estimate:

$$\widehat{\text{Bias}}_{jack} = (n-1)(\hat{\theta}_{jack} - \hat{\theta})$$

Bias-corrected estimator:

$$\hat{\theta}_{JK} = n\hat{\theta} - (n-1)\hat{\theta}_{jack}$$

2. Bayesian Foundations

Core Formula

$$\pi(\theta \mid x) = \frac{p(x \mid \theta)\pi(\theta)}{p(x)} \propto p(x \mid \theta)\pi(\theta)$$

- $\pi(\theta)$ = prior
- $p(x|\theta)$ = likelihood
- $\pi(\theta|x)$ = posterior

Mini Example (Bernoulli)

$$p(x|\theta) = \theta^x(1-\theta)^{1-x}$$

Uniform prior: $\pi(\theta) = 1$

Posterior:

$$\pi(\theta|x) \propto \theta^x(1-\theta)^{1-x}$$

3. Conjugate Priors

Beta–Binomial

$$X|p \sim \text{Binomial}(n,p), \quad p \sim \text{Beta}(\alpha,\beta)$$

Posterior:

$$p|x \sim \text{Beta}(\alpha + x,\ \beta + n - x)$$

Posterior mean:

$$\mathbb{E}[p|x] = \frac{\alpha + x}{\alpha + \beta + n}$$

Mini Example

Prior: Beta(2,2), Data: $x = 7, n = 10$

Posterior: Beta(9,5)

Normal–Normal

$$X_i|\mu \sim N(\mu,\sigma^2), \quad \mu \sim N(\mu_0,\tau^2)$$

Posterior:

$$\mu|x \sim N(\mu_n,\tau_n^2)$$

$$\tau_n^2 = \frac{1}{n/\sigma^2 + 1/\tau^2}$$

$$\mu_n = \frac{n\bar{x}/\sigma^2 + \mu_0/\tau^2}{n/\sigma^2 + 1/\tau^2}$$

4. Model Selection (AIC, BIC)

AIC

$$\text{AIC} = -2\ell(\hat{\theta}) + 2k$$

BIC

$$\text{BIC} = -2\ell(\hat{\theta}) + k \log n$$

Interpretation

- AIC $\to$ prediction (KL divergence)
- BIC $\to$ model identification

5. Penalization (Ridge & Lasso)

Ridge

$$\min_{\beta} \sum (y_i - x_i^T \beta)^2 + \lambda \sum \beta_j^2$$

Lasso

$$\min_{\beta} \sum (y_i - x_i^T \beta)^2 + \lambda \sum |\beta_j|$$

6. Cross-Validation & Overfitting

Key Idea

$$\mathbb{E}[\text{Training Error}] < \mathbb{E}[\text{Test Error}]$$

K-fold CV

$$CV(K) = \frac{1}{K} \sum_{k=1}^{K} \text{MSE}_k$$

LOOCV

$$CV(n) = \frac{1}{n} \sum_{i=1}^{n} (y_i - \hat{y}_{(i)})^2$$

Final Summary Box

- Bootstrap approximates sampling distributions using $\hat{F}_n$
- Jackknife gives bias corrections via leave-one-out
- Bayesian inference: Posterior $\propto$ Likelihood $\times$ Prior
- Conjugacy = same family after updating
- AIC/BIC balance fit and complexity
- CV estimates test error; training error is optimistic